Skills Summary

Compositions of Rigid Motions and Symmetry

Use this reference while completing your worksheet or when studying.

Skill 1 — Doing a composition one step at a time

Coordinate rules. Reflection in the x-axis: $(x, y) \to (x, -y)$. Reflection in the y-axis: $(x, y) \to (-x, y)$. Reflection in $y = x$: $(x, y) \to (y, x)$. Translation by $\langle h, k\rangle$: $(x, y) \to (x + h,\ y + k)$. Rotation about the origin: 90° counterclockwise $(x, y) \to (-y, x)$; 180° $(x, y) \to (-x, -y)$; 90° clockwise $(x, y) \to (y, -x)$.

Order matters. Apply the first transformation to the original, then apply the second to *that* image. Swapping the order usually changes the answer, so write A' down before you go looking for A''.

Example: Reflect $B(5, 2)$ in the y-axis, then translate by $\langle 1, -3\rangle$.
Step 1. Two named steps in a stated order, so use the coordinate rule for each in turn rather than trying to see the finished position on the grid.
Step 2. Reflection in the y-axis: $B' = (-5, 2)$. Then add the vector: $x = -5 + 1$ and $y = 2 + (-3)$.
$\Rightarrow$ $B'' = (\mathbf{-4},\ \mathbf{-1})$

Try it: Reflect $D(-2, 6)$ in the x-axis, then translate by $\langle 4, 1\rangle$.

check: $D'' = (\mathbf{2},\ \mathbf{-5})$

Skill 2 — Naming the single transformation

Two rigid motions done one after the other always add up to a single rigid motion, and it has a one-step name.

Two reflections in PARALLEL mirrors = a translation, perpendicular to the mirrors, of *twice* the distance between them, in the direction from the first mirror to the second.

Two reflections in INTERSECTING mirrors = a rotation about the point where they cross, through *twice* the angle between them, turning from the first mirror towards the second. Perpendicular mirrors therefore give a half turn.

Naming one is not finished until you have said the kind and, for a translation, how far and which way, or, for a rotation, the angle, the direction and the centre.

Example: Reflect in $y = 1$, then in $y = 5$. Name the single transformation.
Step 1. The two mirrors are parallel, so the answer is a translation — and the fastest way to pin it down is to track one convenient point.
Step 2. Take the origin: reflecting in $y = 1$ gives $(0, 2)$, and reflecting that in $y = 5$ gives $(0, 8)$. The mirrors are 4 apart, and the point moved twice that.
$\Rightarrow$ a translation of **8** units straight up, $\langle 0, 8\rangle$.

Try it: Reflect in $x = 2$, then in $x = 5$. Name the single transformation.

check: a translation of **6** units right

Skill 3 — Line and rotational symmetry

A figure has a **symmetry** when a rigid motion maps it exactly onto itself.

Line symmetry: a fold line with the figure matching itself on both sides. **Rotational symmetry:** a turn about the centre that lands the figure back on itself. Its *order* is how many such turns there are in a full revolution, and the smallest angle is $360° \div$ order.

A regular n-gon has exactly n lines of symmetry and rotational symmetry of order n. A parallelogram that is not a rectangle or rhombus has *no* line of symmetry but still has order 2: the half turn about the common midpoint of its diagonals.

Example: What is the smallest angle of rotational symmetry of a regular pentagon?
Step 1. Its five vertices are equally spaced around the centre, so the smallest turn that works is the one carrying each vertex to its neighbour — one fifth of a full turn.
Step 2. $360 \div 5$, and the pentagon also has 5 lines of symmetry, one through each vertex and the midpoint of the opposite side.
$\Rightarrow$ **72** degrees, with rotational symmetry of order 5.

Try it: What is the smallest angle of rotational symmetry of a regular octagon?

check: **45** degrees, order 8